Brier is being built in the open for the Meta Global AI Developer Hackathon. Follow the build →

THE DECISION LAYER FOR ROBOT CI

Know the result before you run it.

Brier predicts every simulated rollout on a pull request, runs only the ones that can change the answer and tells you what it skipped.

Request access | See a sample check

Simulation only · no hardware needed to start

Pull request #412 · Tighten gripper release timing

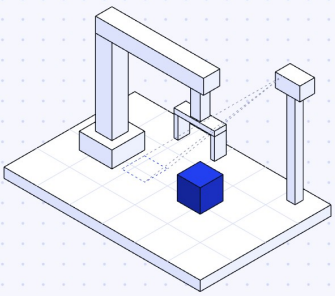
brier / robot-ci — BLOCK

Ran 6 of 16 rollouts. 2 regressions confirmed.

Expected missed regressions ≤ 5% · 10 rollouts skipped

SCENARIO	P(REGRESS)	DECISION	RESULT
aloha-s03	0.91	RUN	BLOCK — release fires ~2 steps early
aloha-s11	0.78	RUN	BLOCK — grip slips during lift
pusht-s14	0.63	RUN	PASS
aloha-s05	0.41	RUN	PASS
pusht-s02	0.12	SKIP	not run
aloha-s07	0.04	SKIP	not run

Failure frame — aloha-s03



SAM 3.1 mask on the step where the cube was released.

Sample data

BUILT ON: Muse Spark | Muse Glimmer | SAM 3.1 | LeRobot | PushT | ALOHA

THE PROBLEM

Simulation CI is the new bottleneck.

Every policy change fans out into dozens of rollouts, each one minutes to hours of GPU. Most come back the same as last time.

ONE CHANGE, SIXTEEN SCENARIOS

Run everything — 16 of 16 rollouts

All GPU time. Nothing missed.

Sample by hand — 5 of 16 rollouts

Cheaper. Unknown misses.

Brier — 6 of 16 rollouts

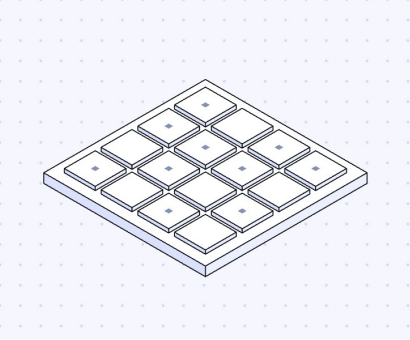
Only the uncertain ones. ≤ 5% expected misses.

Illustrative suite

HOW IT WORKS

Predict. Decide. Run.

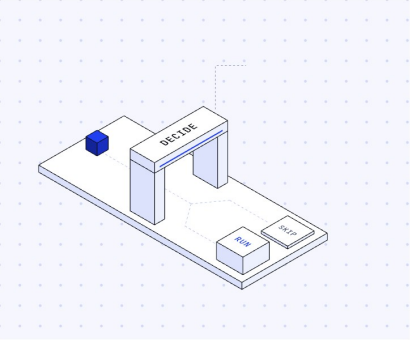
Three models sit between your pull request and your simulator.



STEP 01

Predict

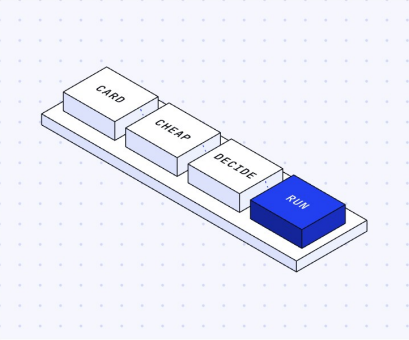
A world model forecasts what each rollout will report.



STEP 02

Decide

A calibrated model prices the risk and picks RUN or SKIP.



STEP 03

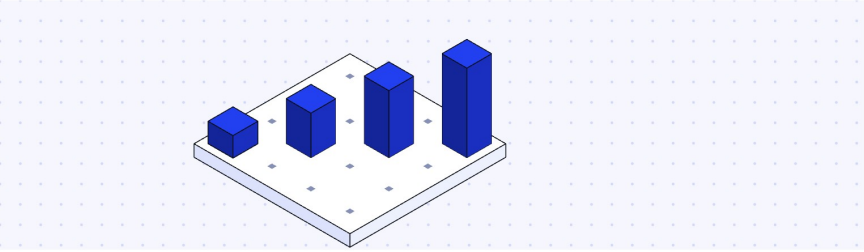
Run and explain

Only uncertain rollouts run. Failures come back masked.

FEATURES

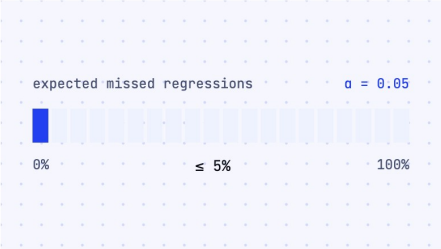
Everything a skip needs to be trusted.

Each decision carries a probability, a bound and a way to check it.



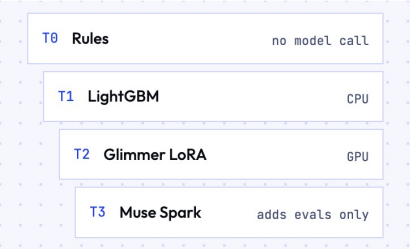
Calibrated probabilities

Trained on log loss and Brier score. When Brier says 0.9, the rollout fails nine times in ten.



A bound on every skip

Conformal risk control keeps expected missed regressions at or below 5%.



Cheapest model first

Most decisions stop at rules or LightGBM. Muse Spark only sees the hard ones.



Escalates when unsure

Close calls run the full evaluation instead of guessing.



Learns from its skips

A small share of skips still run, so the model keeps checking itself.

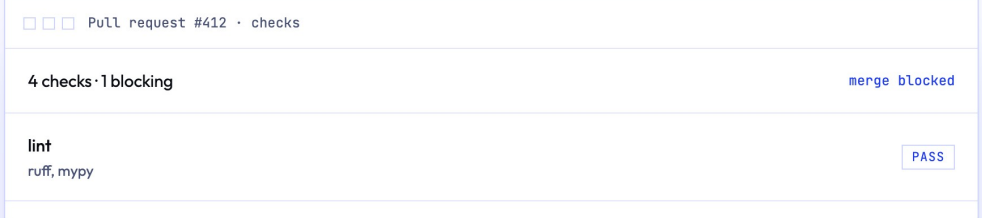
WHERE IT RUNS

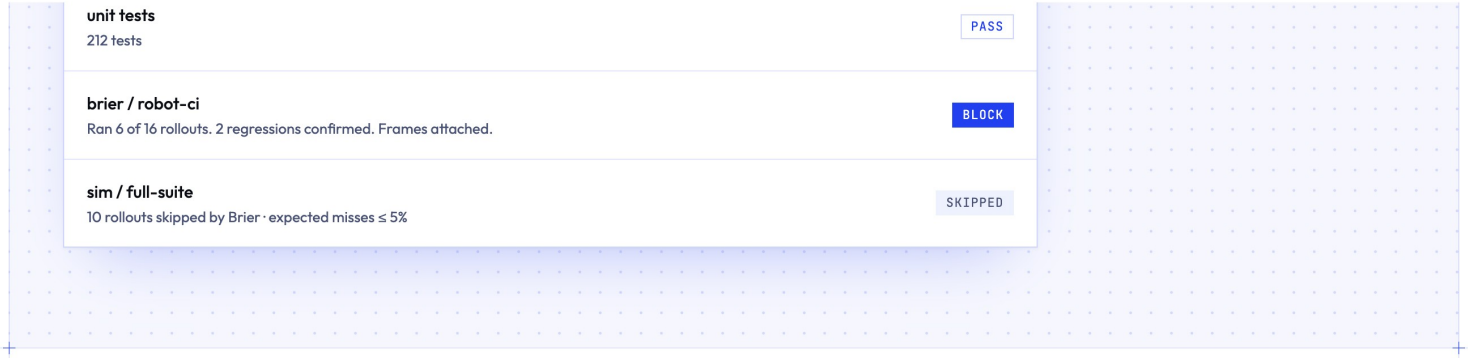
In the pull request, or anywhere you call it.

GitHub check first. CLI and API share the same decision engine.

GitHub check | CLI | Python API

One check on every pull request, with the verdict, probabilities and failure frames.





COMPARE

Cheaper than running it all. Safer than guessing.

How teams handle simulation CI today, side by side.

	Run everything	Sample by hand	Brier
GPU SPENT PER CHANGE	All of it	Whatever you pick	**Only on uncertain rollouts**
CHANCE OF MISSING A REGRESSION	None	Unknown	**≤ 5% expected, stated up front**
WHY A ROLLOUT FAILED	Logs	Logs	**Masked failure frame on the PR**
GETS BETTER OVER TIME	No	No	**Learns from every rollout it runs**

THE GUARANTEE

Skip with a bound, not a hunch.

≤ 5%
expected missed regressions

0
GPU minutes on a skipped rollout

4
model tiers, cheapest first

ROADMAP

Robots first. Every slow loop next.

One decision engine. One adapter per loop.

NOW

Robot CI

PushT and ALOHA suites in LeRobot, as a GitHub check.

NEXT

Software CI

The same engine on test suites for agent-written code.

LATER

Your loop

Any evaluation that is slow, costly and mostly predictable.

FAQ

Questions, answered.

Anything else, ask us when you request access.

What is a calibrated decision model? -

A model whose probabilities match what happens. Of all the rollouts it scores 0.9, about 90% regress. Brier trains one on proper scoring rules and uses it to choose RUN, SKIP or ESCALATE.

What does "≤ 5% expected missed regressions" mean? +

Do I need a robot? +

Which models does it use? +

Can the reasoning model skip evaluations? +

Is it open source? +

EARLY ACCESS

Spend compute where the answer is unknown.

Robotics teams running simulation suites can request access now.



Request access

Star on GitHub

brier

Calibrated decisions for expensive engineering loops.

PRODUCT

- How it works
- Features
- Compare
- Roadmap

DEVELOPERS

- GitHub
- Docs
- Decision plugin

COMPANY

- Request access
- FAQ
- X

© 2026 Brier

Built for the Meta Global AI Developer Hackathon

brier

YOUR WORKSPACE

See every decision before it ships.

Runs, scenarios and the guarantee for every policy change.

Explore the demo runs first >

Sign in

Sign in or create a workspace.

Email address

you@company.com

Continue with email

or

Continue with Google

Continue with GitHub

Use a passkey

By continuing you agree to the [TERMS] and [PRIVACY POLICY].

Need a hand? Read the getting started guide

STEP 2 OF 4

Set up your first suite

About 10 minutes. Calibration then runs in the background.

✓ **Install the GitHub App**

Installed on northbay/manipulation and northbay/sim-policies.

2 **Connect a suite**

Pick an adapter and point Brier at your simulation config.

Robot CI
LeRobot simulation suites: PushT, ALOHA

Software CI NEXT
Test suites and CI jobs

```
# brier.yaml, committed to northbay/manipulation
suite: aloha-transfer-cube
env: gym_aloha/AlohaTransferCube-v0
policy: policies/aloha.yaml
scenarios:
  pose_bins: 4
  lighting: [normal, low-light]
  jitter: [off, on]
tiers:
  cheap: { episodes: 3, horizon: 0.5 }
  expensive: { episodes: 20 }
```

Validate config | Use a template

3 **Calibrate**

Brier runs perturbed versions of your policy in simulation to learn λ. Until then it runs everything.

4 **Open a pull request**

Your first Brier check appears on the next policy change.

WHAT HAPPENS NEXT

01 Brier generates a change library from your policy: noise, latency, camera shifts and more.

02 It runs both tiers on every scenario to label the library.

03 It fine-tunes the decision model and sets λ for your α.

04 From then on, each pull request gets a calibrated check.

Calibration cost: about [GPU-HOURS] on your runners.

brier

Northbay Robotics
Team plan

- Overview
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Docs

Jordan Lee
jordan@northbay.dev

Good morning, Jordan

Northbay Robotics · last 14 days

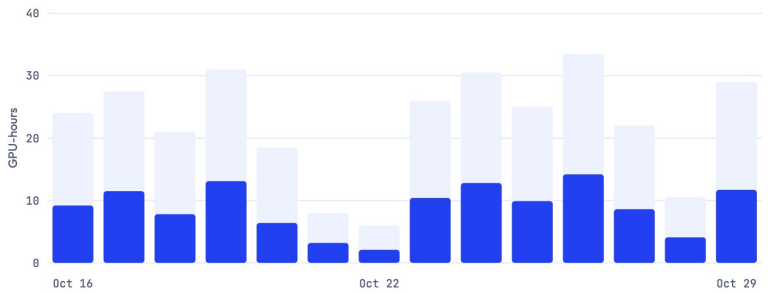
7d | 14d | 30d

2 blocked pull requests — Review →
chg-0418 and chg-0417 have confirmed regressions.

ALOHA insertion is still calibrating — Details →
Brier runs every scenario until 64 changes are labelled.

CHANGES EVALUATED	REGRESSIONS CAUGHT	GPU-HOURS SAVED	MISSED (ESTIMATED)
148	23	196	3.2%
+23 vs previous 14 days	on 19 blocked changes	61% of running every scenario	from exploration runs · bound 5%

GPU-hours per day



USAGE THIS PERIOD

148 of [PLAN LIMIT]
evaluated changes

Expensive evals	1,012
Cheap-tier runs	2,368
Muse Spark calls	87

Billing and usage

Suites

Add suite

SUITE	STATUS	Λ	CALIBRATED	MODEL
ALOHA transfer cube LeRobot · gym_aloha	Guarantee holding	0.18	2 days ago	glimmer-ft-v3
PushT LeRobot · gym_pusht	Guarantee holding	0.21	2 days ago	glimmer-ft-v3
ALOHA insertion LeRobot · gym_aloha	Calibrating · 41 of 64 changes	–	running	glimmer-ft-v3

Recent runs

All runs →

CHANGE	VERDICT	EXPENSIVE EVALS	DATE
Re-mount wrist camera chg-0418 · PR #3	BLOCK	5 / 16	Oct 31, 14:22
Move inference to edge box chg-0417 · PR #2	BLOCK	6 / 16	Oct 31, 11:05
Bump logging, no policy change chg-0416 · PR #1	PASS	0 / 16	Oct 31, 09:40
Retune gripper gains chg-0415 · PR #41	PASS	3 / 16	Oct 30, 18:12
New camera driver chg-0414 · PR #40	PASS	2 / 16	Oct 30, 16:57

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Docs

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Runs

Every policy change Brier evaluated, newest first.

All | PushT | ALOHA

Guarantee active Expected missed regressions ≤ 5% across changes — α 0.05 λ 0.18 calibrated on 64 changes model glimmer-ft-v3

EVALUATIONS	BLOCKED	EXPENSIVE EVALS	EVAL-HOURS SAVED
12	6	52 / 192	67%
policy changes, last 3 days	regression confirmed on an expensive run	scenarios run on the expensive tier	vs running every scenario

Evaluations

latest 50

CHANGE	ENV	FAMILY	VERDICT	EXPENSIVE EVALS	HOURS SAVED	DATE
Re-mount wrist camera chg-0418 · PR #3	ALOHA	camera_shift	BLOCK	5 / 16	61%	Oct 31, 14:22
Move inference to edge box chg-0417 · PR #2	ALOHA	action_latency k=2	BLOCK	6 / 16	55%	Oct 31, 11:05
Bump logging, no policy change chg-0416 · PR #1	PushT	benign	PASS	0 / 16	100%	Oct 31, 09:40
Retune gripper gains chg-0415 · PR #41	ALOHA	action_scale 1.1	PASS	3 / 16	74%	Oct 30, 18:12
New camera driver chg-0414 · PR #40	PushT	obs_noise σ=0.03	PASS	2 / 16	80%	Oct 30, 16:57
Shorter action chunks chg-0413 · PR #39	PushT	chunk_len 8	BLOCK	7 / 16	49%	Oct 30, 13:31
Lighting rig swap chg-0412 · PR #38	ALOHA	brightness	PASS	4 / 16	68%	Oct 30, 10:08
Checkpoint from overnight run chg-0411 · PR #37	ALOHA	weight_noise	BLOCK	9 / 16	36%	Oct 29, 22:46
Calibration offset fix chg-0410 · PR #36	PushT	action_bias	PASS	2 / 16	80%	Oct 29, 19:15
Frames drop under load chg-0409 · PR #35	ALOHA	frame_drop	BLOCK	6 / 16	55%	Oct 29, 16:03
Refactor obs pipeline chg-0408 · PR #34	PushT	benign	PASS	0 / 16	100%	Oct 29, 12:27
Cable in camera view chg-0407 · PR #33	ALOHA	occlusion	BLOCK	8 / 16	43%	Oct 29, 10:51

Runs / chg-0417

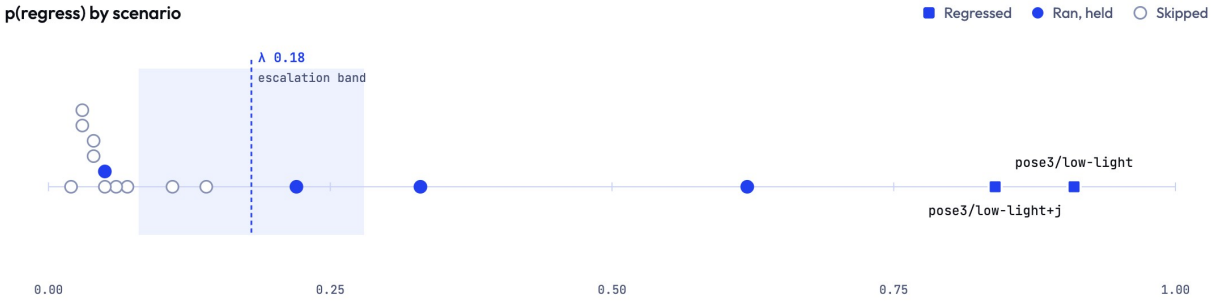
Move inference to edge box

ALOHA transfer cube · action_latency k=2 · PR #2 in northbay/manipulation · Oct 31, 11:05

View pull request | Run full suite

VERDICT	EXPENSIVE EVALS	REGRESSIONS	HOURS SAVED	GUARANTEE
BLOCK	6 of 16	2 confirmed	55%	misses ≤ 5% · λ 0.18

p(regress) by scenario



T0 not benign → A cheap tier on 16 → T1 9 confident → T2 7 scored → T3 +1 eval → B 6 expensive → ✓ BLOCK

Scenarios

16 scenarios · sorted by p(regress)

SCENARIO	P(REGRESS)	DECIDED BY	ACTION	RESULT (SUCCESS RATE)
aloha/pose3/low-light	0.91	T2 Glimmer	RUN	REGRESSED 0.86 → 0.40
aloha/pose3/low-light+j	0.84	T2 Glimmer	RUN	REGRESSED 0.82 → 0.45
aloha/pose4/low-light	0.62	T2 Glimmer	RUN	HELD 0.88 → 0.80
aloha/pose2/low-light+j	0.33	T2 Glimmer	RUN	HELD 0.90 → 0.86
aloha/pose4/normal+j	0.22	T3 Muse Spark	RUN	HELD 0.92 → 0.89
aloha/pose2/normal	0.14	T3 Muse Spark	SKIP	not run
aloha/pose1/low-light	0.11	T3 Muse Spark	SKIP	not run
aloha/pose4/normal	0.07	T2 Glimmer	SKIP	not run
aloha/pose3/normal	0.06	T2 Glimmer	SKIP	not run
aloha/pose1/low-light+j	0.05	T1 LightGBM	SKIP	not run
aloha/pose1/normal+j	0.05	T1 LightGBM	RUN · EXPLORE	HELD 0.94 → 0.93
aloha/pose2/low-light	0.04	T1 LightGBM	SKIP	not run
aloha/pose3/normal+j	0.04	T1 LightGBM	SKIP	not run
aloha/pose4/low-light+j	0.03	T1 LightGBM	SKIP	not run
aloha/pose2/normal+j	0.03	T1 LightGBM	SKIP	not run
aloha/pose1/normal	0.02	T1 LightGBM	SKIP	not run

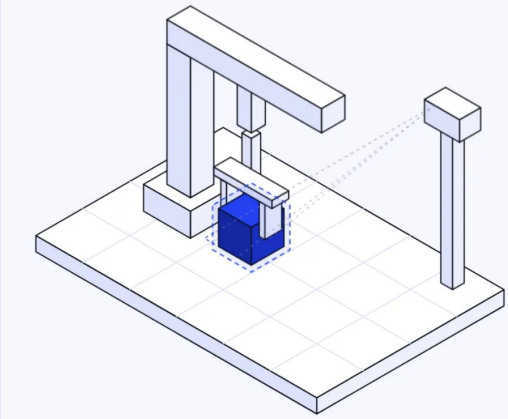
Failures

Why `aloha/pose3/low-light` failed

REGRESSED success 0.86 → 0.40

frame f7 of 8 · release

SAM 3.1 overlay | Raw frame



f1 f2 f3 f4 f5 f6 f7 f8

FAILURE MODE

Misalignment

CAUSE HYPOTHESIS

Actions arrive about 2 steps late, so the gripper opens before the arm reaches the target.

EVIDENCE

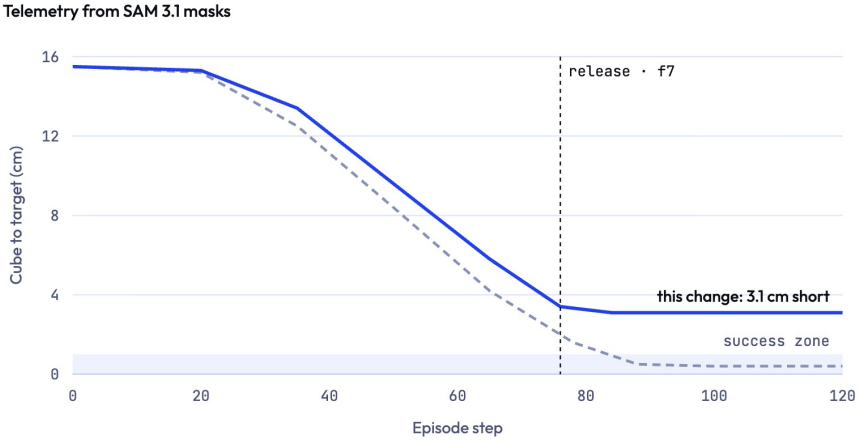
Frames f6 and f7. Cube offset from target 3.1 cm at release.

SUGGESTED CHECK

Latency compensation in the controller

Explained by Muse Spark from keyframes, SAM 3.1 masks and telemetry. The change card was not shown to it.

Telemetry from SAM 3.1 masks



`aloha/pose3/low-light+j` REGRESSED 0.82 → 0.45 Same failure mode under physics jitter. Short in 11 of 20 episodes. Open →

Docs

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brier

Northbay Robotics
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- Docs

Jordan Lee
jordan@northbay.dev

Scenarios

The fixed seeds and conditions each change is evaluated on.

Edit brier.yaml

ALOHA transfer cube | PushT | ALOHA insertion

SCENARIOS
16
4 poses × 2 lighting × 2 jitter

EXPENSIVE TIER
20 episodes
per scenario · stands in for hardware

CHEAP TIER
3 episodes
7.5% of an expensive run

MOST SKIPPED
pose2/normal
skipped on 7 of the last 10 changes

Last 10 changes per scenario

Regressed · Ran, held · Skipped

SCENARIO	BASELINE SUCCESS	LAST 10 CHANGES (NEWEST RIGHT)	SKIP RATE	EXPENSIVE RUN
aloha/pose1/normal	0.94		30%	6.5 min
aloha/pose1/normal+j	0.91		40%	8.5 min
aloha/pose1/low-light	0.88		30%	8.0 min
aloha/pose1/low-light+j	0.85		10%	10.0 min
aloha/pose2/normal	0.94		70%	6.5 min
aloha/pose2/normal+j	0.91		50%	8.5 min
aloha/pose2/low-light	0.88		60%	8.0 min
aloha/pose2/low-light+j	0.85		50%	10.0 min
aloha/pose3/normal	0.90		20%	6.5 min
aloha/pose3/normal+j	0.87		50%	8.5 min
aloha/pose3/low-light	0.84		0%	8.0 min
aloha/pose3/low-light+j	0.81		50%	10.0 min
aloha/pose4/normal	0.94		10%	6.5 min
aloha/pose4/normal+j	0.91		20%	8.5 min
aloha/pose4/low-light	0.88		20%	8.0 min
aloha/pose4/low-light+j	0.85		30%	10.0 min

Models

How well calibrated the decision models are, from brier replay on the test split.

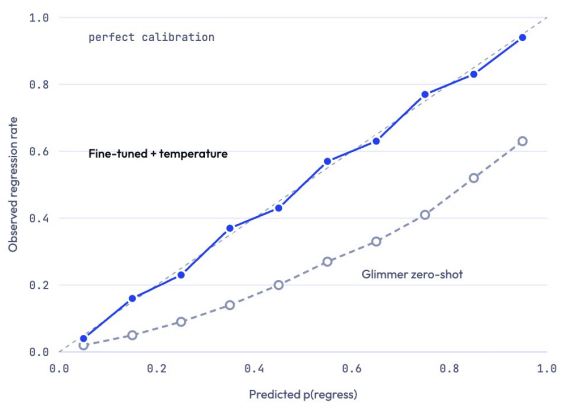
report.json | Weights on Hugging Face

Robot CI | Software CI PHASE 2

ECE	EVAL-HOURS SAVED	GUARANTEE HOLDS	EXPLAINER TOP-2
0.031	58%	19 / 20	76%
fine-tuned + temperature · target ≤ 0.05	full cascade at α = 0.05	splits with realized misses ≤ α	cause family from frames only

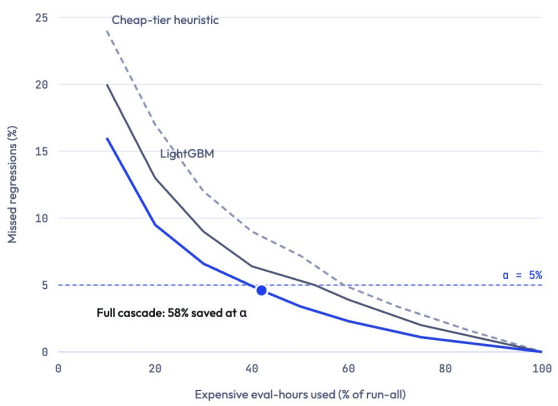
Reliability diagram

Regressed question, 10 equal-mass bins shown



Missed regressions vs eval-hours

Lower and further left is better



Ablation

test split · 16 changes · 256 pairs

STRATEGY	ECE	AUROC	EVAL-HOURS SAVED	CHANGE RECALL	GUARANTEE HOLDS	OOD ECE	P95 LATENCY
Run all expensive	–	–	0%	100%	yes	–	–
Random at equal budget	–	0.50	58%	61%	0 / 20	–	<1 ms
Cheap-tier heuristic	–	0.78	41%	92%	17 / 20	–	<1 ms
T1 LightGBM	0.071	0.86	47%	95%	18 / 20	0.112	2 ms
Glimmer base, zero-shot	0.162	0.71	22%	93%	14 / 20	0.188	410 ms
Glimmer frozen head	0.058	0.87	49%	96%	18 / 20	0.094	380 ms
Glimmer fine-tuned	0.044	0.91	55%	97%	19 / 20	0.077	380 ms
Fine-tuned + temperature	0.031	0.91	56%	97%	19 / 20	0.069	380 ms
Full cascade (T0 to T3)	0.029	0.92	58%	98%	19 / 20	0.066	420 ms
3B student (stretch)	–	–	–	–	–	–	–

Failure explainer

Muse Spark with SAM 3.1 masks, scored against the injected cause

FAILURE MODE	TOP-1	TOP-2	FAILURES
Grasp or drop failure	71%	86%	21
Misalignment	64%	79%	28

REPORT

Model	glimmer-ft-v3
Temperature	1.37
λ at α = 0.05	0.18

Docs

JL
Jordan Lee
jordan@northbay.dev

Stall or timeout	58%	73%	12
Collision	50%	67%	6
Oscillation	44%	61%	9

Calibration changes	64
OOD families held out	occlusion, chunk_len
Generated	Oct 31, 08:15

brier

N Northbay Robotics
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jordan@northbay.dev

Settings

Northbay Robotics

- General
- Guarantee policy
- Repositories
- Members
- Billing and usage
- API keys
- Integrations

Guarantee policy

Applies to every suite in Northbay Robotics. Changes take effect on the next pull request.

Missed-regression bound (α)
Brier picks the threshold so expected missed regressions stay at or below this.

1% | 5% | 10%

Exploration rate (ε)
Share of skipped scenarios that still run, so the miss rate stays measurable.

5 % of skips

Escalation band
Scenarios this close to the threshold go to Muse Spark, which can only add evaluations.

0.10 ± around λ

Expensive budget per change
Hard cap on expensive evaluations. Above it, Brier blocks and asks for a full run.

10 of 16 scenarios

Recalibrate
Recompute λ when enough new labelled changes land in the ledger.

10 new changes

Run everything
Pause decisions and run every scenario on every pull request.

Cancel | Save policy

brier

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404

This page landed off the diagonal.

We predicted a page here. We were wrong, and we logged it.

Back to overview | Home

Docs

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